

Joshua Cook - ENGINEERING PHYSICS

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SKILLS

Hardware & Process Engineering: KiCad (4-Layer PCB, EMI), Micro-Soldering Rework (Fine-Pitch Jumpers), SMT Reflow, Thermal Profiling, Closed-Loop PID Control, Fusion 360

Digital Logic & Firmware: 7400-Series Discrete Logic, Computer Architecture (ISA & Microcode), Logic Simulation, C/C++, STM32, I2S, SPI, DMA Pipelines

Instrumentation & Software: DMM, Oscilloscopes, Logic Analyzers, Python (PySide/Qt, Telemetry), Git, Linux

EXPERIENCE AND TECHNICAL PROJECTS

Audio 2 Bluetooth PCB

Apr. 2026 – Present

- Engineered a 4-layer custom PCB using KiCad around an STM32F4 microcontroller, integrating an audio ADC and an HCI Bluetooth transceiver for wireless audio transmission.
- Diagnosed and resolved board-level hardware flaws by performing micro-soldering rework, including IC orientation corrections, fine-pitch trace jumpers, and pin isolation.

Reflow toaster oven

Jun. 2026 – Jul. 2026

- Implemented a digital Proportional-Integral-Derivative (PID) control algorithm in C++ to minimize overshoot and maintain accurate temperature tracking throughout heating cycles.
- Developed a PySide/Qt desktop interface in Python to visualize real-time thermal telemetry and send dynamic setpoint parameters to the microcontroller via serial communication.
- Calibrated PID loop parameters and designed a custom leaded solder (Sn63/Pb37) reflow profile, ensuring reliable component adhesion while preventing thermal damage.

8-bit Breadboard Computer

Oct. 2024 – Jun. 2025

- Architected an 8-bit breadboard computer using discrete 7400-series logic gates, EEPROMs, registers, and SRAM, incorporating memory-mapped I/O with 8 GPIO pins.
- Simulated the full CPU architecture in a logic gate simulator prior to hardware assembly to validate clock timing, bus control signals, and instruction cycles.
- Designed a flexible EEPROM-based microcode instruction decoder, enabling custom instruction set architecture (ISA) modifications directly through software firmware updates.

Painter

Jun. 2026 – Aug. 2026

Magellan Coatings, Squamish

- Collaborated within multi-disciplinary teams of 2–5 workers across 13+ job sites to complete surface restoration, renovation, and new construction finishing projects.

PhotoLab and Tech Specialist

Aug. 2024 – Sep. 2025

London Drugs, Squamish

- Diagnosed and maintained high-volume printing and scanning equipment during live operations, minimizing machinery downtime through proactive troubleshooting and repair.

EDUCATION

Engineering Physics, BAsC. (in-progress)

UBC, Vancouver